Monday Lab

Q1 Imagine you are tasked with developing a program to manage student information. Each student has a name and is associated with marks in three different subjects. Your goal is to calculate the percentage for each student and identify those whose percentage falls below a user-defined threshold.

Test Data :

Input the string :

Enter the number of students: 3

Enter details for Student 1:

Name: ram

Enter marks for three subjects:

Subject 1: 45

Subject 2: 60

Subject 3: 55

Enter details for Student 2:

Name: shyam

Enter marks for three subjects:

Subject 1: 80

Subject 2: 78

Subject 3: 84

Enter details for Student 3:

Name: monu

Enter marks for three subjects:

Subject 1: 78

Subject 2: 67

Subject 3: 70

Output:

Student Information:

Name	Subject 1	Subject 2	Subject 3	Percentage
ram	45.00	60.00	55.00	53.33%
shyam	80.00	78.00	84.00	80.67%
monu	78.00	67.00	70.00	71.67%

Enter the percentage threshold: 70

Students with a percentage below 70.00%:

ram 53.33%

Q2.Write a C program to create a singly linked list and insert a new node at a specified position. The program should take the number of nodes and their data as input, then take the position and value for the new node. Finally, display the linked list.

Example : Input the number of nodes: 3

Input data for node 1: 10

Input data for node 2: 20
Input data for node 3: 30
Input the position to insert the new node: 2
Input data for the new node: 15
Output: The updated linked list is: 10 -> 15 -> 20 -> 30

Q3 Write a C program to create a singly linked list and delete a node at a specified position. The program should take the number of nodes and their data as input, then take the position of the node to be deleted. Finally, display the updated linked list.
Example: Input the number of nodes: 4
Input data for node 1: 10
Input data for node 2: 20
Input data for node 3: 30
Input data for node 4: 40
Input the position of the node to delete: 3

Output: :The updated linked list is: 10 -> 20 -> 40

Q4 Write a C program to create a file called example.txt, write the sentence "Hello, File Handling!" into it, and then read and display the content of the file.
Example:
Content of the file example.txt:
Hello, File Handling!

Q5. Write a C program to create a file named `data.txt`. The program should take a user's name and age as input, store them in the file, and then read and display the content of the file.

Input your name: John
Input your age: 25

The content of the file data.txt is:
Name: John, Age: 25